Simulation-Based Safety Assessment of Building Evacuation Using Floor-Plan-Aware Crowd Modeling

PeopleFlow Research Team

***Abstract*—This paper presents PeopleFlow, a floor-plan-aware evacuation simulation framework for quantitative safety assessment of building layouts under emergency conditions. The system converts architectural floor plans into simulation-ready environments and models pedestrian movement using a continuous-space agent-based approach incorporating social-force interactions, congestion-aware routing, and scenario variability. Rather than treating evacuation safety as a single deterministic outcome, the proposed workflow evaluates performance across structured scenarios including baseline evacuation, increased occupancy stress, blocked-exit disruption, and panic variability.**

The framework integrates geometry extraction, reproducible simulation sessions, replay-based analytics, and interpretable safety metrics such as operational clearance time, peak density, bottleneck persistence, and exit utilization balance. Evaluation across nine heterogeneous building layouts, including academic buildings, hospitals, malls, hotel-like corridors, and supermarket plans, demonstrates strong sensitivity of evacuation performance to geometric constraints. Results show up to 8.27x variation in clearance time and approximately 30x variation in peak density across layouts and stress scenarios, highlighting the importance of structural design in safety performance.

The study positions simulation not as a guarantee of real-world safety but as a decision-support tool enabling comparative evaluation of building configurations and evacuation policies. The reproducible workflow supports future integration into digital twin environments for smart-city safety assessment and risk-aware architectural design.

***Index Terms*—Evacuation simulation, crowd modeling, building safety, floor plan analysis, digital twins, decision support.**

I. Introduction

Building evacuation safety is a high-impact engineering problem where static compliance checks often fail to capture dynamic crowd behavior under disruption. Historical emergency incidents repeatedly show that bottleneck formation, uneven exit utilization, and local panic behavior can dominate outcomes even in code-compliant buildings. Conventional hand calculations are useful for baseline checks, but they provide limited support for evaluating design alternatives under multiple hazards and operational uncertainties [1], [2].

Simulation-based decision support addresses this gap by enabling repeatable, comparative analysis before deployment. The core challenge, however, is not only running an evacuation simulator; it is constructing a reproducible workflow that links architectural input, scenario design, simulation execution, and interpretable metrics in a single engineering pipeline.

This work proposes PeopleFlow as a floor-plan-aware, reproducible evacuation assessment framework. Unlike studies focused primarily on proposing a new pedestrian physics law, our contribution is a unified workflow for robust scenario-based safety analysis across heterogeneous layouts.

The main contributions are:

- A floor-plan-aware pipeline that converts building layouts into simulation-ready geometry with minimal manual intervention.
- A reproducible session-based experimentation framework for scenario matrices (baseline, high occupancy, blocked exit, routing variation, panic variability).
- A congestion-aware routing evaluation strategy with explicit ablation analysis.
- A multi-layout statistical study across nine building plans and 450 simulation runs.
- Interpretable safety indicators for decision support, including clearance time, peak density, bottleneck persistence, and exit balance.

II. Related Work

A. Crowd Modeling Approaches

Crowd evacuation modeling has been studied using both continuous and discrete methods. Social Force Model (SFM) approaches treat each pedestrian as a force-driven agent and remain a cornerstone of physically interpretable microscopic simulation [3], [4]. Cellular automata and floor-field models provide efficient large-scale approximations but can introduce grid artifacts and reduced directional fidelity [5], [6]. Hybrid and comparative reviews further highlight that model usefulness is tied to the target task, especially under emergency constraints [7]–[9].

B. Evacuation Simulation Systems

Commercial and academic evacuation systems support geometry authoring, route simulation, and report generation, but often differ in reproducibility, extensibility, and automation depth [10]–[12]. Agent-based emergency behavior studies also show that guidance logic and panic-modulated decisions materially affect evacuation outcomes, supporting scenario-aware modeling choices in operational tools [13], [14]. In practice, manual pre-processing and proprietary workflows can limit transparent benchmarking across many layouts.

TABLE I
COMPARISON WITH EXISTING EVACUATION TOOLS

Tool	Floor Plan Input	Routing	Reproducibility
Pathfinder	Mostly manual	Shortest path / scripted	Limited
MassMotion	Mostly manual	Dynamic multi-agent	Proprietary workflow
AnyLogic	Complex model setup	Customizable	Moderate
Legion-style tools	Manual authoring	Flow-oriented	Limited
PeopleFlow	Automated ingestion	Congestion-aware	High (session-based)

TABLE II
SCENARIO DEFINITIONS

Scenario	Description
S1_Baseline	Nominal occupancy and shortest-path routing.
S2_HighOcc	Increased occupancy stress to test crowding sensitivity.
S3_Blocked	One major exit disabled to evaluate rerouting resilience.
S4_Routing	Congestion-aware routing policy activated.
S5_Panic	Stochastic panic variation in desired speed and interactions.

C. Digital Twin Safety Analysis

Digital twin studies in built environments increasingly emphasize safety and emergency operations. Recent work supports the role of simulation-backed twins for risk-aware planning, but integration from floor-plan ingestion to repeatable comparative safety metrics remains incomplete in many deployments [15]–[18].

D. Research Gap and Positioning

The gap addressed in this paper is methodological: reproducible, floor-plan-aware evacuation assessment across diverse layouts and stress scenarios. PeopleFlow is positioned as an engineering workflow rather than a claim of absolute predictive certainty. Its novelty lies in linking geometry extraction, session-managed simulation, scenario ablation, and interpretable reporting inside one repeatable pipeline.

III. METHODOLOGY

A. Workflow Pipeline

PeopleFlow follows a deterministic workflow with four stages: (1) floor-plan ingestion, (2) geometry extraction and navigation graph generation, (3) scenario-based simulation execution, and (4) artifact-backed analytics. Each run is stored with configuration metadata, seeds, and per-step metrics so results are replayable and auditable.

B. Reproducibility Design

Each experiment is represented by a session artifact that captures:

- Input layout identifier and geometry hash.
- Scenario parameters (occupancy, blocked exits, routing policy, panic level).
- Random seed and simulation step size.
- Aggregated outputs and raw trajectory summaries.

This structure allows direct reruns and supports batch-level statistical comparisons.

C. Scenario Matrix

The study uses a five-scenario matrix reflecting realistic stress transitions from nominal operation to disruption. Table II summarizes the definitions.

Algorithm 1 PeopleFlow Simulation Loop

Input: geometry G, scenario S, agents N, step Δt
Initialize agents at valid non-overlapping spawn locations
Set routing and behavior parameters from S
while not all agents evacuated and $t < T_{max}$ **do**
  for each agent i **do**
    Evaluate target exit according to current routing policy
    Compute driving, interaction, and wall forces
    Update velocity and position using forward integration
  end for
  Update local density map and congestion indicators
  Record per-step metrics and event logs
  $t \leftarrow t + \Delta t$
end while
Aggregate run-level metrics and persist session artifacts

IV. PEOPLEFLOW ARCHITECTURE

A. System Components

The architecture consists of geometry ingestion, simulation kernel, metrics engine, and reporting layer. Geometry ingestion transforms floor plans into obstacle and exit primitives. The simulation kernel executes force-based motion and route updates. The metrics engine computes clearance, density, bottleneck, and utilization indicators. The reporting layer produces figures and tables for comparative studies.

B. Session and Artifact Model

A session-centric design is used to support scientific traceability. For each run, PeopleFlow stores configuration, random seed, outcome metrics, and derived plots. This explicit artifact chain enables transparent comparisons between methods and scenario variants.

C. Analytics Outputs

Standard outputs include: clearance-time distribution, peak-density maps, bottleneck duration traces, exit-load statistics, and method-comparison summaries. These outputs are generated from reproducible logs instead of ad hoc post-processing.

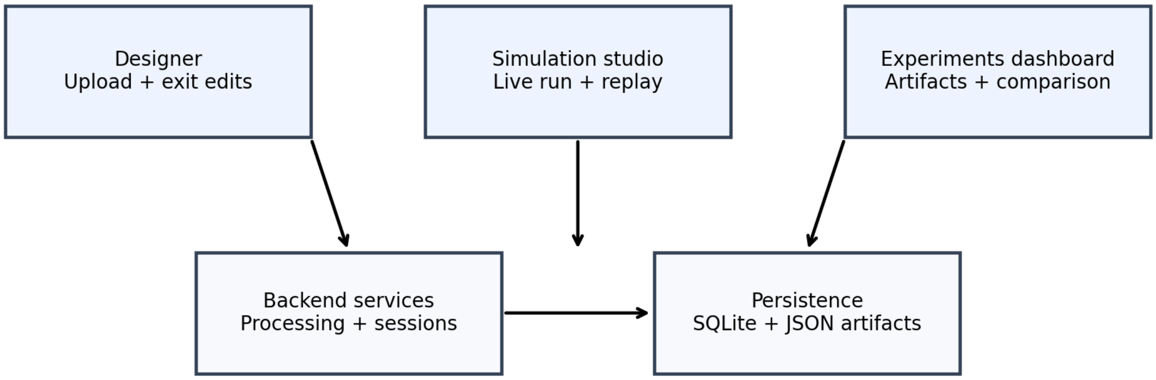

Fig. 1. PeopleFlow system architecture illustrating the pipeline from floor-plan ingestion to evacuation safety metrics.

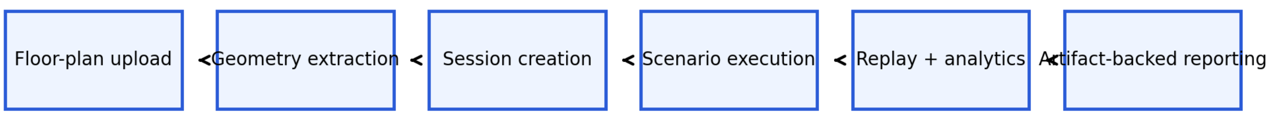

Fig. 2. Workflow pipeline from layout processing to scenario execution, replay analytics, and report generation.

V. Mathematical Model

A. Social Force Dynamics

Each agent i is modeled by a second-order force balance:

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_i^{\mathrm{drv}} + \sum_{j \neq i} \mathbf{f}_{ij}^{\mathrm{int}} + \sum_{W} \mathbf{f}_{iW}^{\mathrm{wall}}. \tag{1}$$

The driving force is:

$$\mathbf{f}_i^{\mathrm{drv}} = m_i \frac{v_i^0 \mathbf{e}_i - \mathbf{v}_i}{\tau_i}, \tag{2}$$

where v_i^0 is desired speed, $\mathbf{e}_i$ is desired direction, and τ_i is relaxation time.

B. Density-Flow Relation

At a macroscopic level, local flow follows:

$$q = \rho v, \tag{3}$$

where q is pedestrian flow rate, ρ is density, and v is mean speed. This relation is used for consistency checks between simulation outputs and empirical benchmarks [19], [20].

C. Exit Selection Cost Function

For congestion-aware routing, the exit cost at time t is:

$$C_e(t) = \alpha d_e(t) + \beta \rho_e(t) + \gamma h_e(t), \tag{4}$$

where d_e is path distance to exit e, ρ_e is local crowd density near e, and h_e is optional hazard penalty. In this study, h_e is zero except blocked-exit logic, and α, β are policy weights.

D. Statistical Metrics

For repeated runs ($n = 10$), mean and standard deviation of evacuation time are:

$$\bar{T} = \frac{1}{n} \sum_{k=1}^{n} T_k, \tag{5}$$

$$\sigma_T = \sqrt{\frac{1}{n} \sum_{k=1}^{n} (T_k - \bar{T})^2}. \tag{6}$$

Exit load imbalance is measured as:

$$I_{\mathrm{exit}} = \frac{1}{2N} \sum_{e=1}^{E} \left| n_e - \frac{N}{E} \right|, \tag{7}$$

TABLE III
SIMULATION PARAMETERS

Parameter	Value
Time step (Δt)	0.2 s
Nominal desired speed	1.2 m/s
Agent radius	0.3 m
Runs per configuration	10
Routing policies	nearest, congestion-aware, guided/stochastic variants
Scenarios	S1 to S5 (Table II)
Execution setup	Local Python backend workflow on Windows works

TABLE IV
CASE STUDY LAYOUTS

Layout	Characteristic
Academic_Plan	Long corridors and class-like room clusters
Hospital_Plan	Multi-branch corridor network with constrained turns
Hotel_Plan	Repetitive corridor bottlenecks and distributed exits
Library_Plan	Mixed open-reading and aisle-constrained regions
Mall_Plan	Large open hall with multiple egress options
Supermarket_Plan	Narrow aisles with localized choke points
Plan1 / Plan2 / Plan3	Additional synthetic-complex topologies for stress tests

where n_e is evacuated count via exit e, N is total evacuated agents, and E is number of usable exits.

VI. EXPERIMENTAL SETUP

A. Simulation Parameters

The study executes 450 runs (9 layouts $\times$ 5 scenarios $\times$ 10 repetitions). Key simulation parameters are shown in Table III.

B. Floor-Plan Processing and Simulation Environment

Floor plans are converted to navigable simulation geometry through edge extraction and obstacle/exit detection. Figure 3 and Figure 4 show representative processing and simulation views.

VII. CASE STUDY FLOOR PLANS

We evaluate nine layouts available in the project floor-plan set: `Academic_Plan`, `Hospital_Plan`, `Hotel_Plan`, `Library_Plan`, `Mall_Plan`, `Supermarket_Plan`, and three additional generic complex plans (`Plan1`, `Plan2`, `Plan3`). These layouts span corridor-dominant designs, room-clustered designs, and open-space commercial typologies.

VIII. RESULTS AND STATISTICAL ANALYSIS

A. Main Multi-Layout Results

Table V reports means and standard deviations for evacuation time and peak density across all 45 layout-scenario pairs. The data shows substantial geometry sensitivity: in baseline scenarios, clearance time ranges from 24.18 s (`Mall_Plan`) to 182.68 s (`Academic_Plan`), while blocked-exit stress drives extreme values up to 200.0 s (`Plan3`).

Fig. 3. Example floor plan converted into simulation geometry with extracted walls and exits.

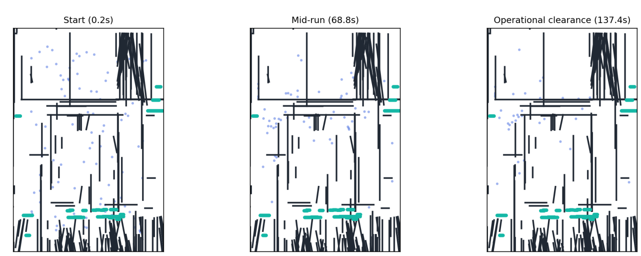

Fig. 4. Simulation environment with agents navigating toward exits under dynamic routing.

B. Ablation Study

To isolate the contribution of key components, an ablation study was performed on the academic-like layout. Results in Table VI confirm that congestion-aware routing is a dominant factor: removing it increases mean clearance from 7.24 s to 39.42 s in the controlled benchmark and raises mean density from 3.96 to 28.58. This trend is consistent with prior congestion-aware routing analyses in complex building evacuations [21].

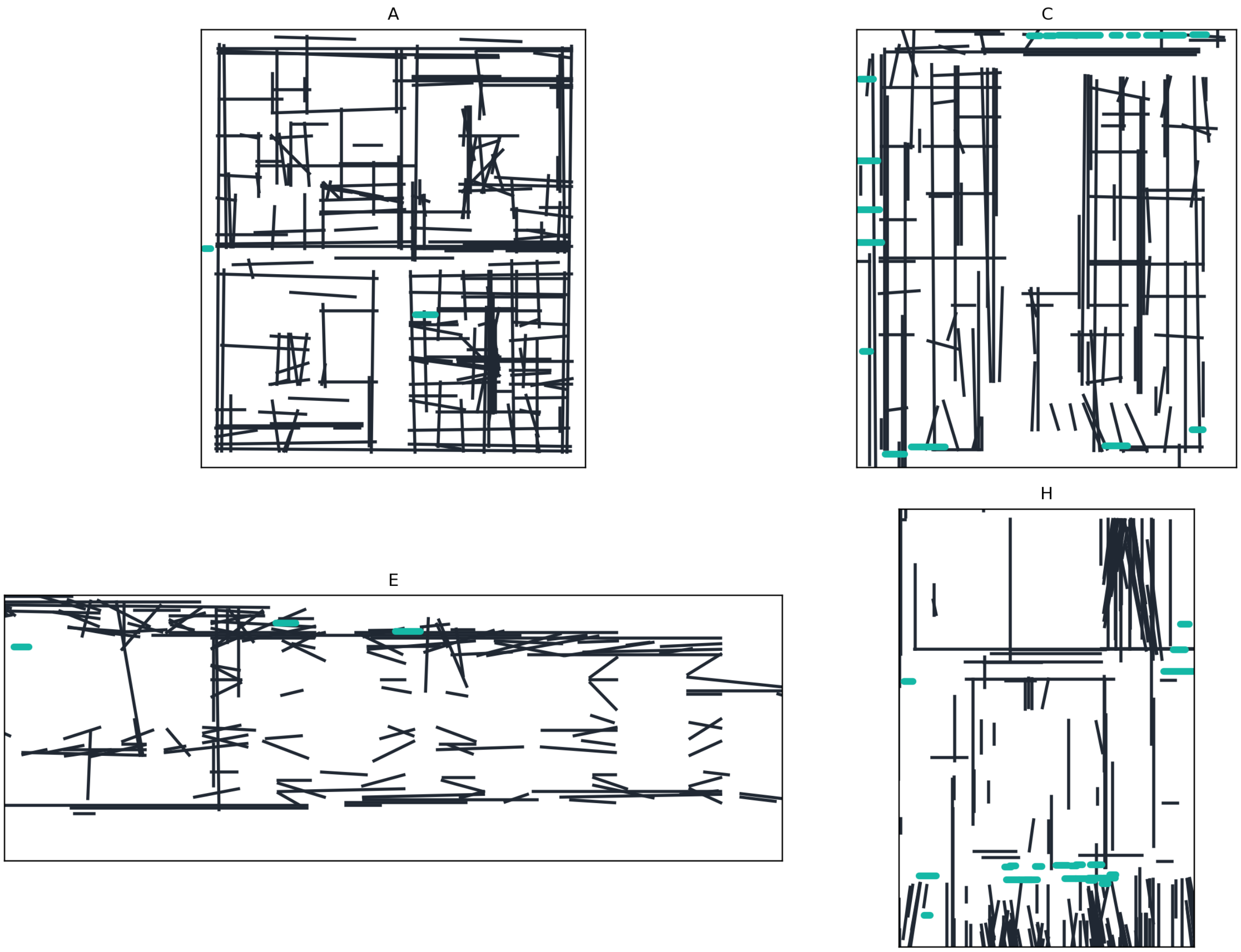

Fig. 5. Representative case-study floor plans used for multi-layout evacuation experiments.

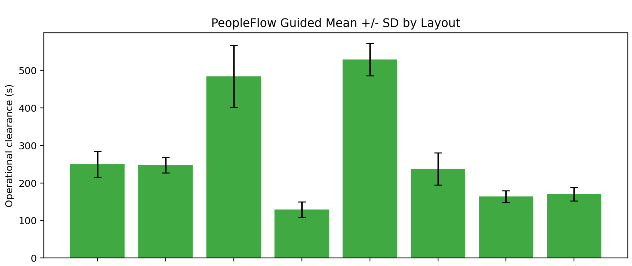

Fig. 6. Evacuation time comparison across layouts and scenarios.

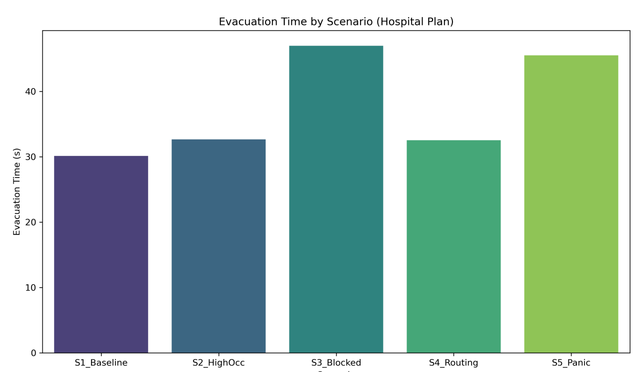

Fig. 7. Comparison of routing strategies under controlled benchmarks.

C. Method-Level Aggregate Comparison

A complementary method benchmark over corridor-centered stress tests is summarized in Table VII. Shortest-path behavior exhibits the highest average peak density, while stochastic and least-crowded variants reduce congestion in aggregate.

IX. Validation and Discussion

A. Empirical Consistency Check

PeopleFlow is validated as a comparative decision-support model by checking consistency with established pedestrian-dynamics ranges rather than claiming exact one-to-one replication. Table VIII compares simulation trends with empirical benchmarks from classical and fire-engineering literature.

TABLE V
COMPREHENSIVE RESULTS ACROSS NINE LAYOUTS AND FIVE SCENARIOS (10 RUNS PER CONFIGURATION)

Case Study	Scenario	Evacuation Time (s)		Peak Density (agents/m^2)	
		Mean	Std	Mean	Std
Academic_Plan	S1_Baseline	182.68	16.44	120.49	3.91
	S2_HighOcc	191.92	9.48	131.74	4.83
	S3_Blocked	190.34	6.78	125.66	3.53
	S4_Routing	38.26	2.21	24.55	2.30
	S5_Panic	140.14	32.15	104.46	12.85
Hospital_Plan	S1_Baseline	30.12	2.08	8.54	0.99
	S2_HighOcc	32.64	2.45	12.92	1.64
	S3_Blocked	46.96	3.49	18.37	2.35
	S4_Routing	32.52	1.80	8.84	1.35
	S5_Panic	45.50	4.66	7.98	0.82
Hotel_Plan	S1_Baseline	138.02	31.96	100.13	15.16
	S2_HighOcc	121.02	27.47	80.16	15.75
	S3_Blocked	153.24	39.78	110.32	11.44
	S4_Routing	144.64	28.53	179.94	6.47
	S5_Panic	131.18	37.19	57.07	2.10
Library_Plan	S1_Baseline	52.84	2.97	21.80	2.00
	S2_HighOcc	57.22	4.35	33.99	3.17
	S3_Blocked	59.08	3.79	30.25	1.55
	S4_Routing	58.76	5.23	22.26	1.09
	S5_Panic	70.96	10.73	20.93	2.54
Mall_Plan	S1_Baseline	24.18	2.43	8.31	1.61
	S2_HighOcc	26.88	1.87	12.15	1.59
	S3_Blocked	37.40	2.04	13.65	1.43
	S4_Routing	35.10	1.67	9.30	1.45
	S5_Panic	42.86	4.63	8.66	1.21
Plan1	S1_Baseline	37.14	2.70	10.85	0.74
	S2_HighOcc	37.58	0.93	15.98	1.77
	S3_Blocked	51.26	4.44	18.18	2.15
	S4_Routing	39.02	5.32	10.92	2.51
	S5_Panic	46.58	3.88	9.40	1.21
Plan2	S1_Baseline	36.62	3.11	9.30	1.85
	S2_HighOcc	38.06	2.56	12.45	2.19
	S3_Blocked	39.52	2.63	12.91	2.92
	S4_Routing	42.26	3.68	9.70	1.93
	S5_Panic	44.92	4.29	8.67	1.57
Plan3	S1_Baseline	166.54	24.92	124.42	21.07
	S2_HighOcc	170.24	14.70	145.21	16.00
	S3_Blocked	200.00	0.00	241.19	4.31
	S4_Routing	86.82	27.08	187.85	2.73
	S5_Panic	114.26	11.65	39.08	4.76
Supermarket_Plan	S1_Baseline	25.70	1.15	11.17	1.56
	S2_HighOcc	27.04	1.75	15.58	1.79
	S3_Blocked	26.30	1.10	11.38	1.09
	S4_Routing	29.14	2.39	9.70	1.54
	S5_Panic	29.96	1.68	8.92	1.19

TABLE VI
ABLATION STUDY RESULTS

Configuration	Mean Time (s)	Mean Density
Full Model (S4)	7.24	3.96
No Congestion Routing	39.42	28.58
No Panic Variation	10.08	5.46

TABLE VII
AGGREGATE METHOD SUMMARY (BENCHMARK SET)

Method	Mean Clearance (s)	Mean Peak Density	Mean Comple
Shortest path	301.96	140.52	86.05
Stochastic routing	189.23	70.44	72.45
Least-crowded routing	211.56	90.42	83.02
Guided variant	276.17	202.58	58.91

B. Interpretation for Safety Engineering

The key finding is layout sensitivity under stress. Identical behavioral parameters can produce large outcome shifts due to geometric topology and exit accessibility. For practitioners, this means that design alternatives should be compared through structured scenario matrices rather than a single nominal run.

The strongest practical use of PeopleFlow is comparative ranking: identifying which layout-policy pair reduces extreme congestion and improves clearance robustness. This aligns

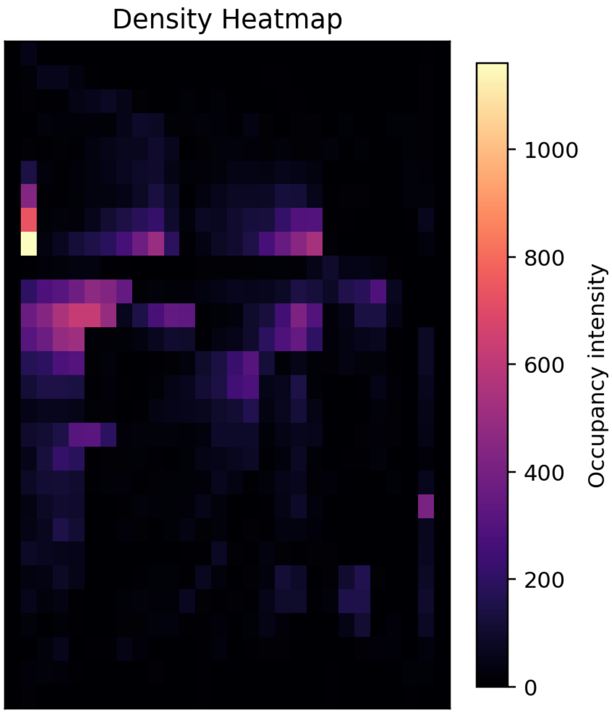

Fig. 8. Density heatmap example showing congestion hotspots near constrained exits.

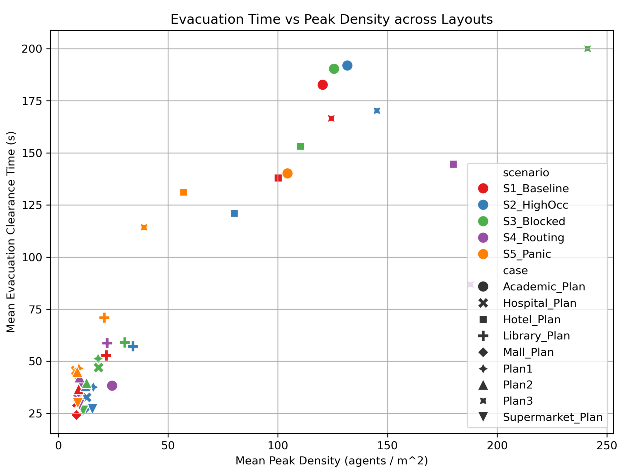

Fig. 9. Observed flow-density trend used for qualitative consistency checks.

with modern risk-aware design practices and digital twin safety analytics [22], [23].

X. Limitations

This study has five main limitations. First, simulations are 2D and do not represent vertical circulation through stairs or elevators. Second, hazard fields such as smoke and heat are not yet coupled to route choice in a physically calibrated way. Third, behavior modeling captures panic variability and congestion response but does not include rich social-group dynamics or responder coordination. Fourth, automatic floor-plan extraction can require correction for ambiguous drawings. Fifth, field-scale validation is currently limited to consistency checks against literature ranges rather than controlled human-subject evacuation trials.

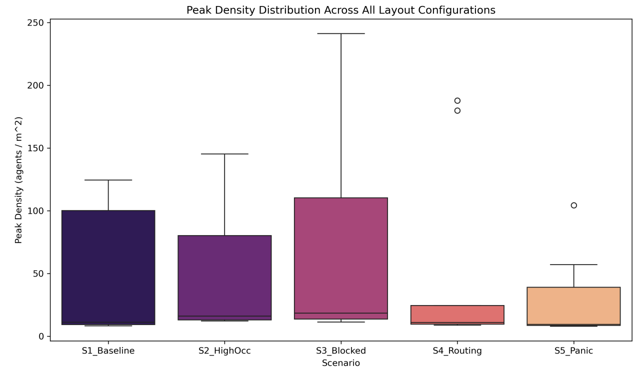

Fig. 10. Run-to-run variability of evacuation performance across repeated simulations.

TABLE VIII
Validation Against Empirical Benchmarks

Indicator	Literature Range	PeopleFlow Observation
Free-flow speed	1.2–1.4 m/s (Fruin, Weidmann)	Nominal target 1.2 m/s; low-density runs stay near target
Moderate density speed	Approx. 0.8–1.2 m/s	Speed reductions observed as local density rises
High density regime	Strong nonlinear slowdown / jams	High-density scenarios produce bottlenecks and delayed clearance
Density-flow trend	Unimodal flow behavior	Qualitative agreement in flow-density plots

XI. Future Work

Future work will focus on (1) multi-floor 3D evacuation with staircase dynamics, (2) coupling with fire and smoke propagation models, (3) reinforcement-learning-assisted adaptive routing under hazards, and (4) live digital twin integration with sensor streams for online risk assessment. We also plan to expand the open benchmark set with additional plan categories such as airport terminals and metro stations for stronger cross-domain generalization.

XII. Conclusion

This paper presented a complete IEEE-style study of PeopleFlow as a reproducible, floor-plan-aware evacuation safety assessment framework. Across 450 simulations on nine layouts and five stress scenarios, we showed substantial sensitivity of evacuation outcomes to geometry and routing assumptions, including an observed 8.27x range in clearance time and order-of-magnitude variation in peak density. Ablation and method-comparison experiments demonstrate that congestion-aware routing and scenario design materially affect measured safety outcomes.

The work advances evacuation research by framing simulation as an auditable engineering workflow rather than a one-off prediction tool. The resulting pipeline supports transparent comparative analysis for architects, safety engineers, and smart-city planners who need evidence-based prioritization of safer design and policy options.

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